

Capstone Progress Report

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Machine Learning Model

We downloaded an initial notebook from Kaggle that had excellent results but he used a different version of the dataset we couldn't find, and when we tried to use his methodology on our original dataset it didn't achieve good accuracy.

We trained our own model and these are the results:

Model	Condition	Accuracy	Charged off Precision	Charged off Recall	Charged off F1
XGBoost	Structured	71.0%	29.0%	56.3%	0.383

Other notebooks on Kaggle that used our exact dataset had really similar results. We are still trying to improve the ML model as best as we can, but it makes us suspect the features are not predictive enough for machine learning.

LLM

In the meantime we ran some experiments with LLMs and we tested 3 models:

- GPT - 5 has strong performance and achieves a higher accuracy than our current ML model

Model	Condition	Accuracy	Charged off Precision	Charged off Recall	Charged off F1
XGBoost	Structured	71.0%	29.0%	56.3%	0.383
Gemini 2.5 Flash	no desc	61.0%	24.4%	68.6%	0.361
Gemini 2.5 Flash	desc	58.0%	19.0%	50.0%	0.276
Gemini 2.5 Pro	no desc	58.0%	24.0%	75.0%	0.364
Gemini 2.5 Pro	desc	57.0%	21.3%	62.5%	0.317
GPT - 5	no desc	73.0%	26.1%	37.5%	0.308
GPT - 5	desc	80.0%	38.9%	43.8%	0.412

Consistency

Since LLMs are not deterministic, we ran GPT-5 three times per condition on the exact same 100 loans to check if the results are reproducible.

Condition	Run 1	Run 2	Run 3	Mean	Std
no desc	79%	77%	79%	78.3%	1.15%
desc	78%	80%	80%	79.3%	1.15%

Condition	Stable (3/3 agree)	Unstable	Stability
no desc	92	8	92%
desc	93	7	93%

Robustness

To make sure our results weren't specific to the 100 loans we picked, we sampled a completely new non-overlapping batch of 100 loans and ran the same experiment.

This batch was harder: 23 Charged Off vs 16 in the original.

- The LLM is less accurate (76% vs. 70%), but the ranking still holds:

GPT-5 desc (76%) > GPT-5 no desc (72%) > XGBoost (70%)

Model	Condition	Original sample	New sample	Difference
XGBoost	Structured	71%	70%	-1 %
GPT-5	no desc	73%	72%	-1 %
GPT-5	desc	80%	76%	-4 %